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Evaluation of variant specific trypanolysis tests for serodiagnosis of human infections with *Trypanosoma brucei gambiense*

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Twelve *T.b. gambiense* clone populations of distinct Variable Antigen Type (VAT) were combined in immune lysis tests with 340 sera of trypanosome infected patients from 8 different African countries and 267 non trypanosomiasis control sera. The diagnostic specificity of the test was 100%. At a serum dilution of 1:4 the overall test sensitivity with single VATs varied from 39.1 to 98.2% and from 12.1 to 86.8% at 1:32. At a serum dilution of 1:32 some combination tests with 2 VATs still scored above 96%. The VAT recognition patterns were clearly correlated with the geographical origin of the sera, reflecting a diversity in variable antigen repertoires.

Key words: Sleeping sickness; *Trypanosoma brucei gambiense*; Serodiagnosis; Variable antigen; Immune trypanolysis

Introduction

Due to the low sensitivity of parasitological techniques, antibody detection tests are currently used to facilitate diagnosis of human infections with *T.b. gambiense*.

Apart from the Procyclic Agglutination Trypanosomiasis Test (Liu et al., 1989; Ngaira et al., 1992) all contemporary assays are making use of antigen preparations derived from bloodstream form trypanosomes (BSF) grown in rodents. BSFs have a large capacity for antigenic variation (Barry et al., 1991). In addition to a multitude of invariable antigens (Le Ray, 1975), differing in cellular location and concentration, they are coated with a monolayer of Variable Surface Glycoprotein (VSG). This VSG, which determines the Variable Antigen Type (VAT) of the individual trypanosome, represents about 10% of the total protein content. It is highly immunogenic and elicits VAT specific protective antibodies with opsonizing, agglutinating and lytic activity.

There is good evidence that selected variable antigens of *T.b. gambiense* can provide powerful serodiagnostic reagents. This principle has been explored in test systems using fixed trypanosomes such as immunofluorescence (Magnus et al., 1978a; Vervoort et al., 1983) and direct agglutination (Magnus et al., 1978b; Vervoort et al.,

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1983) and in test systems using semi-purified VSG preparations such as ELISA (Vervoort et al., 1978; Vervoort et al., 1983; Depla et al., 1992), indirect haemagglutination (Vervoort et al., 1983) and latex agglutination (Büscher et al., 1991).

Since the pioneering work of Gray (1972, 1974) using direct agglutination tests with live trypanosomes, further comparative research on variable antigen repertoires of *T.b. gambiense* based on immune lysis and immunofluorescence tests with clone populations (Magnus et al., 1983) has amply confirmed that stocks of different geographical origin are producing a lot of predominant IsoVATs in common, i.e. serologically similar VATs expressed at an early stage of infection (Lumsden, 1982).

The present investigation is an updated serodiagnostic evaluation of the immune lysis test on human serum samples, using 12 distinct *T.b. gambiense* VATs, including some promising candidates formerly identified (Magnus et al., 1983).

Materials and Methods

Trypanosomes

Mouse adapted bloodstream form clone populations of 12 distinct Variable Antigen Types (VATs) of *T.b. gambiense* and 1 VAT of *T.b. rhodesiense* were used throughout experiments.

T.b. gambiense VAT clones LiTat 1.1 to 1.10 (Lille Trypanozoon antigen types) were derived from stock Eliane, originally isolated from a patient in Ivory Coast in 1952 (Afchain, 1976 ; Van Meirvenne et al., 1977).

T.b. gambiense VAT clones AnTat 11.17 and 11.20 (Antwerp Trypanozoon antigen types) were derived from stock Mba, originally isolated from a Zairese patient in 1974 (Babiker, 1981). AnTat 11.17 and, rather speculatively AnTat 11.20 as well, cross-react with some metacyclic VATs produced by various *T.b. gambiense* stocks (Aerts and Le Ray, unpublished data).

T.b. rhodesiense ETat 1.2R is a human serum resistant VAT of the Edinburgh Trypanozoon Antigen Repertoire ETAR 1 (Lumsden and Herbert, 1975; Van Meirvenne et al., 1976). It served as a putative negative control VAT since it could not be demonstrated in various *T.b. gambiense* stocks of different geographical origin (Magnus et al., 1983). In the latter study it was referred to as R 1/2.

Working trypanosome populations were obtained from mice 3 days after infection with cryopreserved stabilates.

Variant specific rabbit antisera

These antisera were obtained 6–7 days after intravenous inoculation of 10^7 clone trypanosomes (Van Meirvenne et al., 1975), aliquoted in 300 µl volumes and freeze-dried.

Human sera

Sera from 340 patients with parasitologically confirmed *T.b. gambiense* infection had been collected in Congo (50), Equatorial Guinea (25), Gabon (50), Ivory Coast (43), Nigeria (31), Sudan (43), North-West Uganda (50) and Zaire (48). Of the 234 patients whose cerebrospinal fluid had been examined, 73 showed normal leucocyte counts (up to 5 cells per mm³) and normal protein levels (up to 30 mg/dl) whereas 161 had increased values for one or both of these parameters.

Control sera from 267 individuals without clinical, serological or parasitological suspicion of trypanosomiasis included 230 samples from Benin (20 from Atacora and 210 from N.E. Borgou), 17 from Congo (Bouenza) and 20 from Uganda (Moyo).

All samples had been filtered on a 0.22 μm filter (Nucleopore^R), aliquoted in small volumes and freeze-dried.

Complement-rich guinea pig serum (GPS)

VAT specific immune trypanolysis is an antibody and complement mediated phenomenon (Lourie and O'Connor, 1936, 1937). Male guinea pig serum (GPS) has a high complement activity and is an excellent survival medium for salivarian bloodstream form trypanosomes. This makes GPS an ideal diluent for the immune trypanolysis test (Van Meirvenne et al., 1975).

Freshly collected GPS was pooled, aliquoted in 2ml volumes and stored at -70°C . Prior to use, the trypanolytic complement activity of each batch was determined by checkerboard titration of the rabbit antiserum against LiTat 1.3 trypanosomes in combination with various twofold dilutions of GPS in isotonic Tris-Glucose-Saline buffer supplemented with Ca^{++} and Mg^{++} (TSG: 0.115 M Trizma^R Base, 0.065 M NaCl, 0.06 M Glucose, 2.5 mM MgCl_2 and 8.3 mM CaCl_2 , pH 7.5). The end-titre dilution of GPS thus obtained was currently 1:4. In the actual tests GPS was always used in excess, i.e. undiluted.

Comparative serotyping of VAT clones

Clones were cross-tested against all VAT specific antisera (Van Meirvenne et al., 1975). In the homologous combination a full range titration of twofold antiserum dilutions from 1:4 to 1:2048 was carried out. In the heterologous combinations only antiserum dilutions of 1:4, 1:8 and 1:16 were tested.

Fifty microliters of antiserum dilutions in GPS were prepared in V-shaped microplates (Greiner). Loaded plates, sealed with adhesive tape, could be stored at -70°C for several weeks, which considerably facilitated the practical planning of the experiments. Prior to use, the frozen plates were thawed at room temperature.

A trypanosome suspension in GPS, containing about 10^7 organisms/ml, as estimated by the Matching Method (Herbert and Lumsden, 1976), was prepared from heparinized infected mouse blood and 50 μl added to each serum dilution.

After shaking, plates were incubated at room temperature for 90 minutes. Finally, a drop of the reaction mixture was examined under phase contrast microscopy (250x). The highest antiserum dilution, defined as final dilution in the reaction mixture, still causing 50% lysis was arbitrarily considered as the end-point titre.

VAT homogeneity of the clones was evaluated at the 1:32 dilution of the homologous antiserum by checking 1000 trypanosomes and counting the number of surviving heterotypes.

Tests on human sera

Final serum dilutions of 1:4 and 1:32 were tested.

Initially, when applying the above described test procedure for rabbit antisera, an unexpected prozone phenomenon, reflected by failing trypanolysis at the 1:4 dilution, often occurred with sera of trypanosomiasis patients. This was attributed to some

anticomplementary activity and could completely be circumvented by inserting a pre-incubation step with GPS.

In practice, 25 µl of undiluted test serum and 25 µl of a 1:8 dilution in TSG (see above) were pipetted into separate wells of microplates. Preliminary experiments showed that loaded plates could be stored at -70°C for several weeks. Upon testing, 25 µl of GPS were added and plates pre-incubated at room temperature for 30 minutes. Next, 50 µl of a 10^7 /ml suspension of trypanosomes in GPS were added and plates incubated at room temperature for another 90 minutes. Reading was done as described above and 50% lysis or more recorded as positive.

Results

Comparative serotyping of the VAT clones

The 12 *T.b. gambiense* clones were all antigenically distinct. Homogeneity was virtually 100% except for clone AnTat 11.20 which contained about 5 *pro mille* heterotypes.

Diagnostic specificity of the tests

Of the 267 human control sera, 57 were tested at 1:4 and 1:32 against all 12 *T.b. gambiense* VATs. The remaining 210 samples were screened at 1:4 against LiTat 1.3 and LiTat 1.6 only. All these tests were negative.

Diagnostic sensitivity of the tests

The grouped results obtained at a final serum dilution of 1:4 are shown in Table 1.

The VAT specific recognition patterns are complex and to some extent correlate with the origin of the test sera.

The overall sensitivity of the test clearly depends on the VAT used. It varies from 39.1% for LiTat 1.2 to 98.2% for LiTat 1.5 (Table 3).

Four sera, two from Congo (Bouenza) and 2 from Bas-Zaire were negative with all 12 VATs tested. Fifty-six serum samples from Gabon and Ivory Coast gave a positive result with all 12 *T.b. gambiense* VATs. In order to ascertain that the observed lysis was not due to some non specific trypanocidal serum activity, 54 of these sera were correspondingly tested at 1:4 against *T.b. rhodesiense* VAT ETat 1.2R, without any positive result.

The results obtained at a final serum dilution of 1:32 are shown in Table 2. The overall test sensitivity now varies from 12.1% for LiTat 1.4 to 86.8% for LiTat 1.3 (Table 3). Seven sera were negative with all 12 VATs.

In Table 3, the 12 *T.b. gambiense* VATs have been classified in decreasing order of test sensitivities at the 1:4 and 1:32 serum dilution respectively. Also given is the ratio of the sensitivity at 1:32 versus that at 1:4. The relative loss of sensitivity at 1:32 varies from one VAT to another. Some VATs that are scoring high at 1:4, such as LiTat 1.4, 1.7 and 1.8, are strikingly deficient at 1:32. Only LiTat 1.3, 1.5 and 1.6 score above 70% at either serum dilution.

Table 4 shows 8 combinations of 2 VATs yielding a test sensitivity of 98.5% at 1:4, including 2 combinations yielding 96% at 1:32.

With 3 VATs, a sensitivity of 98.8% at 1:4 is obtained in 4 combinations, one of which still yields 97.3% at 1:32.

TABLE 1

Number of positive results obtained with 12 VATs of *T.b.gambiense* and sera of 340 trypanosome infected patients at a dilution of 1:4

Origin of sera		Number of sera tested	LiTat VATs										AnTat VATs	
Country	Area		1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	11.17	11.20
Congo	Couloir, Cuvette, Pool, Bouenza	50*	43	1	48	47	48	48	47	48	1	43	48	17
Equatorial Guinea	Mbini, Kogo	25	24	14	25	23	25	22	24	24	15	25	24	6
Gabon	?	50	50	47	50	48	49	44	48	48	47	46	43	38
Ivory Coast	Daloa, Vavoua	43	43	42	43	42	43	43	43	43	43	41	35	33
Nigeria	Ethiope East	31	18	0	5	26	30	31	0	29	0	28	30	2
Sudan	Tambura, LiRangu	32	29	29	32	29	32	32	31	32	29	29	30	2
	Juba, Redjaf	11	0	0	10	11	11	10	11	11	0	0	10	10
Uganda	Moyo	50	0	0	50	50	50	50	50	50	0	0	2	50
Zaire	Bas-Zaire	48*	43	0	46	43	46	46	46	46	0	46	35	36
Total		340	250	133	309	319	334	326	300	331	135	258	257	194

* 2 serum samples of each of these series negative with all 12 VATs.

TABLE 2

Number of positive results obtained with 12 VATs of *T.b.gambiense* and sera of 340 trypanosome infected patients at a dilution of 1:32

Origin of sera		Number of sera tested	LiTat VATs										AnTat VATs	
Country	Area		1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	11.17	11.20
Congo	Couloir, Cuvette, Pool, Bouenza	50	32	1	48	4	32	36	19	20	1	9	3	15
Equatorial Guinea	Mbini, Kogo	25	22	8	24	1	24	14	15	8	11	17	4	1
Gabon	?	50	36	23	47	4	47	19	26	37	35	13	9	36
Ivory Coast	Daloa, Vavoua	43	31	21	43	5	34	33	9	24	28	8	3	32
Nigeria	Ethiope East	31	16	0	0	0	27	30	0	20	0	9	15	1
Sudan	Tambura, LiRangu	32	19	11	29	1	21	23	5	23	16	2	6	2
	Juba, Redjaf	11	0	0	10	5	9	6	2	3	0	0	1	10
Uganda	Moyo	50	0	0	50	10	42	47	14	19	0	0	0	50
Zaire	Bas-Zaire	48	30	0	44	11	34	35	19	18	0	6	3	35
Total		340	186	64	295	41	270	243	109	172	91	64	44	182

TABLE 3

Ranking of the 12 *T.b.gambiense* VATs in decreasing order of test sensitivity (%) at serum dilutions 1:4 and 1:32

Serum 1:4			Serum 1:32			Sensitivity at 1:32		
VAT	Sensitivity		VAT	Sensitivity		Sensitivity at 1:4		
LiTat	1.5	98.2	LiTat	1.3	86.8	LiTat	1.3	0.95
	1.8	97.4		1.5	79.4	AnTat	11.20	0.93
	1.6	95.9		1.6	71.5	LiTat	1.5	0.80
	1.4	93.8		1.1	54.7		1.6	0.74
	1.3	90.9	AnTat	11.20	53.5		1.1	0.74
	1.7	88.2	LiTat	1.8	50.6		1.9	0.67
	1.10	75.9		1.7	32.1		1.8	0.52
AnTat	11.17	75.6		1.9	26.8		1.2	0.48
LiTat	1.1	73.5		1.10	18.8		1.7	0.36
AnTat	11.20	57.1		1.2	18.8		1.10	0.24
LiTat	1.9	39.7	AnTat	11.17	12.9	AnTat	11.17	0.17
	1.2	39.1	LiTat	1.4	12.1	LiTat	1.4	0.12

TABLE 4

Cumulative sensitivity (%) of combination tests with 2, 3 or 4 VATs (340 serum samples from trypanosome infected patients)

Combination	Sensitivity	
	at 1:4	at 1:32
LiTat 1.3+1.6		96.5
LiTat 1.3+1.5		96.2
LiTat 1.3+1.8		93.8
LiTat 1.5+1.6	98.5	91.2
LiTat 1.1+1.5		89.1
LiTat 1.5+AnTat 11.20		88.8
LiTat 1.1+1.6		86.5
LiTat 1.4+1.5		81.5
LiTat 1.3+1.5+1.6		97.3
LiTat 1.5+1.6+AnTat 11.20		95.3
LiTat 1.1+1.5+1.6	98.8	94.4
LiTat 1.4+1.5+1.6		92.6
LiTat 1.1+1.3+1.5+1.6	98.8	97.9

Finally, one combination of 4 VATs gave sensitivities of respectively 98.8% at 1:4 and 97.9% at 1:32, as was the case with the complete set of 12 VATs.

Discussion

The present results confirm that the immune trypanolysis test with selected VATs is a valuable tool for serodiagnosis of human *T.b. gambiense* infections. As compared with other antibody detection tests the diagnostic specificity of the assay is exception-

ally high. Actually, the absence of a protective lytic activity against human infective trypanosomes in sera of non trypanosomiasis patients is no surprise. An extra indication for good specificity of the positive results are the VAT dependent reactivity patterns of the trypanosomiasis sera themselves.

The overall sensitivity of the test is equally high. Only 4 out of 340 trypanosomiasis sera failed to react with all the 12 VATs used in the used in the experiments. Such serum samples could be used to search for additional VATs to be included in the screening protocol.

The diagnostic principle of the immune lysis test is based on the progressive accumulation of protective serum antibodies to a multitude of VATs that are successively experienced in the course of infection (Van Meirvenne, 1987). For being efficient the VATs to be used as a diagnostic reagent should have surface epitopes in common with VATs that appear at an early stage of infection and that also have a tendency to revert after initial immune clearance, so as to keep the corresponding antibodies at a detectable level. Moreover, to overcome the diversity in variable antigen repertoires and to achieve a broad spectrum test, diagnostically suitable VATs should have a wide range of distribution, i.e. behave as serologically cross-reacting IsoVATs common to different repertoires (Lumsden, 1982). The present results suggest that all these conditions can be fulfilled by a combination of less than 5 VATs. However, an important factor yet to be taken into consideration is the duration of the infection. All the trypanosomiasis sera tested in the present study had been collected during routine field surveys and maybe only few samples were derived from patients with recent infections. Nevertheless, the results obtained for patients with and without cerebrospinal fluid alterations were completely similar (not shown).

Theoretically, the optimal VATs for early case detection would be the metacyclic VATs (mVATs) injected by the tsetse fly. Unfortunately, very little is known about the mVAT serotypes of *T.b. gambiense* and virtually nothing about their IsoVAT distribution in different repertoires. The bloodstream form VATs AnTat 11.17 and possibly also AnTat 11.20, cross-react with mVATs of some *T.b. gambiense* stocks. In the present experiments both VATs performed averagely. Meanwhile, it has been shown that the AnTat 11.17 VSG gene, initially believed to be unique and generally conserved in *T.b. gambiense* (Dai Do Thi et al., 1991), is missing in several stocks of this subspecies from North-West Uganda (Enyaru et al., 1993) and Ivory Coast (Bromidge et al., 1993). Whenever common mVATs become available, their diagnostic potential for spotting very early and advanced infections certainly merits further evaluation.

From Table 1 it is apparent that part of the VAT recognition patterns is correlated with the geographical origin of the test sera. As regards the LiTAR 1 VATs, conspicuous failing reactivities are those of numbers 2 and 9 in Congo and Bas-Zaïre, 2,3,7, and 9 in Nigeria, 1,2,9 and 10 in the Juba/Redjaf area of Sudan and in the neighbouring Moyo area of North-West Uganda. Similar divergencies are seen with AnTat 11.17 and 11.20. All this reflects a complex diversity in VAT repertoires, in accordance with genetic data showing an irregular distribution of particular VSG genes, including the above mentioned absence of AnTat 11.17, the missing of LiTat 1.3 in some stocks from Cameroon, Ivory Coast, Nigeria and Zaïre (Dukes et al., 1992; Bromidge et al., 1993) and the missing of LiTat 1.6 or its IsoVAT partners of the AnTat 1.8 complex in some stocks from Congo and Cameroon (Pays et al., 1983).

As compared with analogous immune lysis experiments on *T.b. rhodesiense* in Uganda (Isharaza and Van Meirvenne, 1990), the selection of *T.b. gambiense* VATs with a broad-spectrum diagnostic reactivity proves relatively easy. This may be partly related to an extra requirement in the more acute *T.b. rhodesiense* situation, where the test should become positive at an even earlier stage of infection. The major obstacle, however, seems to be the much greater diversity within and between variable antigen repertoires of the latter subspecies. The genetic background of the relative restriction and stability of the variable antigen repertoires in *T.b. gambiense* has been partly elucidated (Dero et al., 1987).

By virtue of its high specificity the *T.b. gambiense* immune lysis assay could serve as a reference antibody test, confined to laboratory use. While awaiting tests based on defined invariant antigens, the lysis test also serves to identify candidate variable antigens for being evaluated in more current test systems. In the meantime, when designing or applying an antibody detection test with rough antigen preparations of bloodstream form trypanosomes, the relative contribution of invariable and variable antigens should explicitly be taken into consideration and strict VAT standardization of the working population is a prerequisite for obtaining reproducible results.

A major disadvantage of antibody detection tests is that they cannot discriminate between ongoing and cured infections. Antigen and DNA detection assays are now being developed for this purpose (Liu and Pearson, 1987; Nantulya, 1989; Nantulya et al., 1992).

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